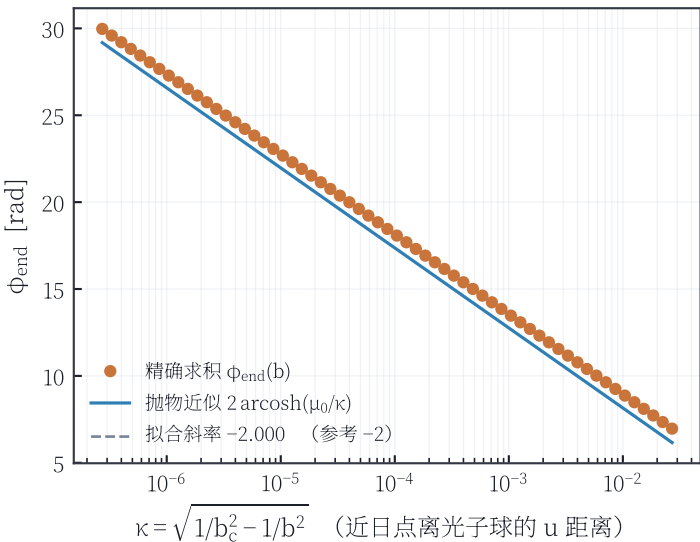
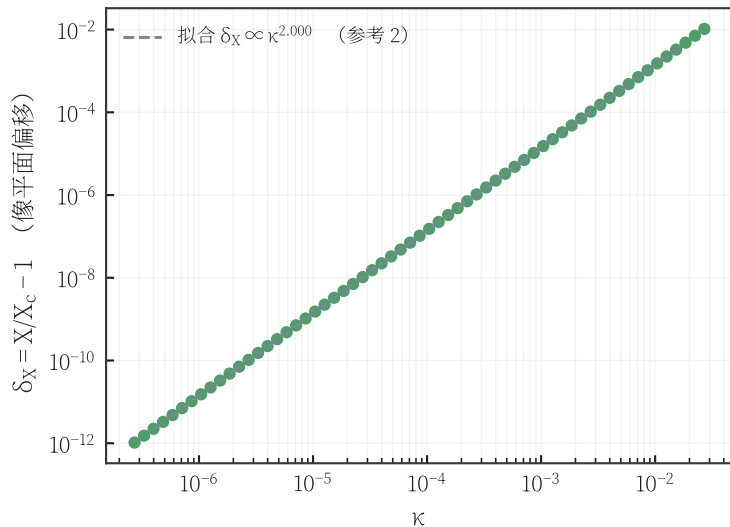


近临界光线的对数发散: $\phi_{\text{end}} \simeq -\ln \kappa$, 像平面斜率为 -1

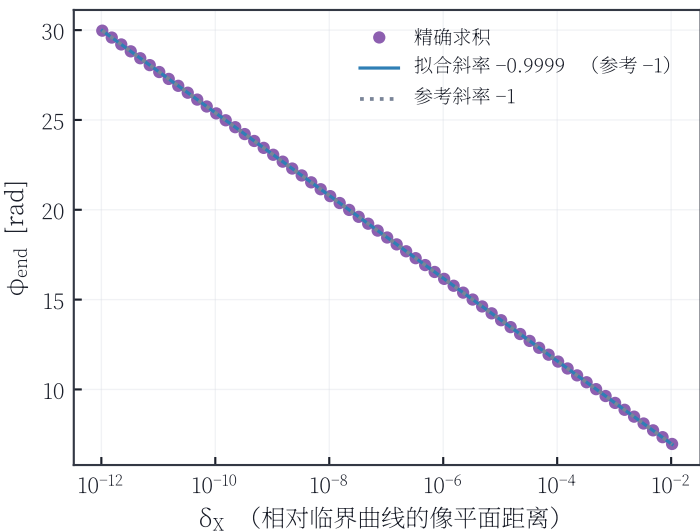
(a) 方位角随 κ 对数发散



(b) 像平面偏移是 κ 的二次映射



(c) 每十倍 δ_X 只多绕 $\ln 10 (= 2.303 \text{ rad})$



(d) 轨道平面内的环绕: 像阶次的来源 (均自观测者出发)

